Cell, Vol. 63, 1179–1191, December 21, 1990, Copyright © 1990 by Cell Press

X:A Ratio, the Primary Sex-Determining Signal in Drosophila, Is Transduced by Helix-Loop-Helix Proteins

Susan M. Parkhurst,*† Daniel Bopp,‡ and David Ish-Horowicz*
* Developmental Genetics Laboratory
Imperial Cancer Research Fund
Developmental Biology Unit
Zoology Department
University of Oxford
South Parks Road
Oxford OX1 3PS
England
‡ Department of Molecular Biology
Princeton University
Princeton, New Jersey 08544

Summary

Drosophila determines its sex by "counting" X chromosomes. We show that premature expression of the pair-rule segmentation gene *hairy* interferes with this process, resulting in female-specific lethality by inhibiting initiation of the master control gene *Sex-lethal* (*Sxl*). The female-specific lethality can be suppressed by a constitutive *Sxl* allele or by extra copies of X-linked "counting elements." These results are best explained by competition between *hairy* and other helix-loop-helix transcription factors that act in chromosome counting. We have confirmed this model by showing that misexpression of the *achaete-scute T4* gene induces ectopic *Sxl* expression and male-specific lethality, confirming that *achaete-scute T4* is the *sisterless-b* counting element. We propose that X chromosomes are counted through heterodimers of helix-loop-helix transcription factors that act synergistically to initiate *Sxl* expression.

Introduction

Sexual reproduction is the predominant source of genetic exchange, variation, and diversification, and has led to the evolution of sexual dimorphism in most eukaryotic organisms. Despite such a widespread and crucial role, sex can be determined by widely different strategies. Sexual fate in mammals and birds is specified by a dominant chromosome: the male-determining Y chromosome in mammals, and the female-determining W in birds (for reviews see Baker and Belote, 1983; Bull, 1983; Baker, 1989; Hodgkin, 1990). In contrast, Drosophila establishes sex according to its "X:A ratio," the ratio between the number of X chromosomes and complete autosomal chromosome sets (Bridges, 1921, 1925). 2X/2A flies are female (X:A = 1); 1X/2A flies are male (X:A = 0.5), independent of the Y chromosome, which is only required for the final stages of sperm maturation in males (for review see Hodgkin, 1990).

† Present address: Division of Biology 156-29, California Institute of Technology, Pasadena, California 91125.

However, superficially diverse modes of sex determination may hide common underlying developmental mechanisms that differ only in their terminal stages. This is best demonstrated in the nematode Caenorhabditis elegans, which determines sex by X:A ratio, as in Drosophila (Madl and Herman, 1979; Hodgkin, 1990). Appropriate mutations in a single gene render its sex dependent on either a single female-determining gene (a WZ female/ZZ male mechanism) or on temperature (environmentally based) (Hodgkin, 1983, 1987a, 1987b).

Sex determination in Drosophila includes two processes: morphological sex differentiation and X chromosome dosage compensation. The former controls the different appearances and behaviors of the two sexes and is propagated through a cascade of sex-specific RNA-splicing events where transcripts are differentially processed in males and females (Cline, 1978, 1979; Maine et al., 1985; Boggs et al., 1987; Bell et al., 1988; Nagoshi et al., 1988; for reviews see Baker, 1989; Hodgkin, 1989, 1990; Steinmann-Zwicky et al., 1990). The latter ensures equal expression of X-linked genes in 1X males and 2X females by hypertranscription of the single X chromosome in males (Lucchesi and Skripsky, 1981; Cline, 1983, 1984; Gergen, 1987). Both processes are regulated through a key X-linked control gene, *Sex-lethal* (*Sxl*), whose activity must be ON in females and OFF in males (Figure 1; Cline, 1978, 1979, 1984, 1985). Inappropriate states of *Sxl* expression (Sxl^{OFF} in females; Sxl^{ON} in males) lead to lethality as a result of aberrant dosage compensation. Initiation of *Sxl* expression depends on X:A ratio, but *Sxl* activity is maintained by differential RNA processing. *Sxl* is transcribed in both sexes, but only female transcripts make full-length active *Sxl* protein. Male transcripts are inactive because they include an extra exon that truncates translation (Bell et al., 1988). *Sxl* expression is under autoregulatory control, as removal of the male-specific exon requires *Sxl* activity. Morphological transformations and intersex phenotypes caused by late perturbations in the sex determination pathway are usually masked by cell lethality due to failed dosage compensation (Sánchez and Nöthiger, 1983; Cline, 1984).

Whereas *Sxl* maintenance is regulated at the level of RNA processing, its initiation appears to be transcriptionally regulated. In addition to an X:A ratio of 1, *Sxl* activation requires the action of a maternal gene, *daughterless* (*da*) (Figure 1; Mange and Sandler, 1973; Cline, 1976, 1980, 1983; Cronmiller and Cline, 1986; Cronmiller et al., 1988). Female progeny of *da* mothers fail to activate *Sxl* and die as embryos, owing to aberrant dosage compensation (Cline, 1980, 1983); male offspring survive, as they do not require *Sxl*. The *da* protein is highly homologous to E12/E47, mammalian transcription factors required for immunoglobulin gene expression, suggesting that *da* is a transcription factor and that *Sxl* initiation may be transcriptionally regulated (Murre et al., 1989a). The earliest *Sxl* transcripts differ in structure from later differentially spliced forms (Bell et al., 1988; Salz et al., 1989) and may provide

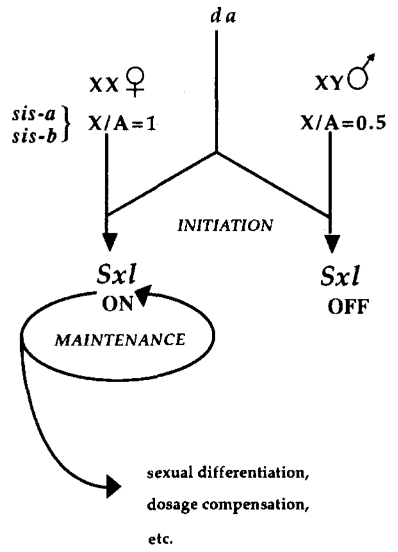

Figure 1. Summary of the Sex Determination Pathway Involving the Master Regulatory Gene, *Sxl*
Simplified from Cline, 1984.

Sxl protein with which to trigger the autoregulatory feedback loop (L. Keyes, T. Cline, and P. Schedl, submitted).

da protein includes a so-called helix-loop-helix (HLH) domain: two linked amphipathic helices preceded by a stretch of basic amino acids that are implicated in protein–protein dimerization and DNA binding (Caudy et al., 1988b; Murre et al., 1989a). The HLH motif is found in several other proteins including *MyoD* and other myogenic control genes and the *T3*, *T4*, *T5*, and *T8* proteins of the Drosophila *achaete-scute* complex (AS-C), which are required for development of the embryonic and adult nervous system (for reviews see Murre et al., 1989a; Benezra et al., 1990). Strikingly, *da* is also required zygotically for development of the embryonic peripheral nervous system (Caudy et al., 1988a; Dambly-Chaudière et al., 1988). Genetic interactions between *da* and AS-C and in vitro DNA binding studies suggest that *da* and AS-C may act through protein heterodimers (Dambly-Chaudière et al., 1988; Murre et al., 1989b). Thus, *da* and AS-C form heterodimers in vitro that bind to the κE2 immunoglobulin gene enhancer, whereas the homodimers bind poorly or not at all. Indeed, it has been suggested that HLH transcription factors may act predominantly as protein heterodimers (Murre et al., 1989b).

As *da* acts maternally in sex determination, it is present in both female and male eggs and cannot define zygotic X:A ratio, which is assessed via dispersed chromosomal sites known as "counting elements" (Dobzhansky and Schultz, 1934; Madl and Herman, 1979; Cline, 1988; McCoubrey et al., 1988; Hodgkin, 1990). Autosomal "denominator" elements that confer male tendency have been postulated but not yet identified (for review see Hodgkin, 1990). However, two X-linked genes, *sisterless-a* (*sis-a*) and *sisterless-b* (*sis-b*), have the characteristics of "numerator" elements for measuring X:A ratio (Cline, 1986, 1988). *sis* mutations reduce effective X:A ratio and promote male development; extra doses of sis^+ predispose to female development and can suppress male biasing mutations (such as *da*) (Cline, 1986, 1988; Torres and Sánchez, 1989).

Whereas *sis-a* is not yet characterized molecularly, *sis-b* maps within a ~20 kb region of AS-C encoding only two identified transcripts, *T4* and *T7*, that are expressed homogeneously during the early cleavage stages, disappearing early in blastoderm stage 14 (Campuzano et al., 1985; Cabrera et al., 1987; Romani et al., 1987; Villares and Cabrera, 1987; Alonso and Cabrera, 1988; Cline, 1988). Recently, it has been shown that mutations at AS-C *T4* affect sex determination, suggesting that the AS-C *T4* gene (also called *scute-α*) contributes significantly to *sis-b* (Torres and Sánchez, 1989).

Our work on spatial and temporal misexpression of the *hairy* (*h*) pair-rule segmentation gene (S. M. P. and D. I.-H., submitted) also implicates HLH transcription factors in Drosophila sex determination. Although *h* protein normally plays no role in sex determination, we found that ectopic *h* expression causes female lethality. *h* encodes an HLH protein that acts in patterning other pair-rule genes and in the development of adult bristle pattern (Ingham et al., 1985a, 1985b; Rushlow et al., 1989). *h* behaves as a negative regulator of *fushi tarazu* (*ftz*) during embryonic segmentation and of *achaete* (AS-C *T5*) during bristle patterning (Falk, 1963; Botas et al., 1982; Moscoso del Prado and Garcia-Bellido, 1984a, 1984b; Carroll and Scott, 1986; Howard and Ingham, 1986; Ish-Horowicz and Pinchin, 1987). As both *h* and AS-C *T5* encode HLH proteins, *h* could interfere with AS-C *T5* activity rather than regulate AS-C expression.

In this paper, we provide in vivo evidence for HLH proteins functioning as heterodimers by describing the effects of HLH protein misexpression. We demonstrate that ectopic *h* prevents initiation of *Sxl* expression by interfering with X chromosome counting. We propose and test a model in which counting elements are X-linked HLH transcription factors that initiate *Sxl* expression as heterodimers with *da*. In particular, we show that AS-C *T4* behaves as a numerator element whose overexpression causes inappropriate *Sxl* activation and male lethality. We discuss roles for HLH heterodimers in sex determination, segmentation, and neural development.

Results

h Expression from *hb-h* Causes Female Lethality

We used the *hunchback* (*hb*) gap gene promoter to examine the effects of misexpressing pair-rule genes in the anterior of the embryo (S. M. P. and D. I.-H., submitted). Here we describe the effects of the *hb-h* fusion gene, in which the *hb* promoter drives ectopic expression of the *h* pair-rule gene. *hb* is first transcribed from nuclear cycle 11 in a broad domain extending over the anterior 45%–50% of the egg, whereas *h* protein is normally expressed at cycle 14 in a series of reiterated stripes (Tautz et al., 1987; Carroll et al., 1988; Schröder et al., 1988; Hooper et al., 1989; Tautz and Pfeifle, 1989). *hb-h* drives *h* expression in the *hb* domain: *h* protein is first detected during nuclear cycle

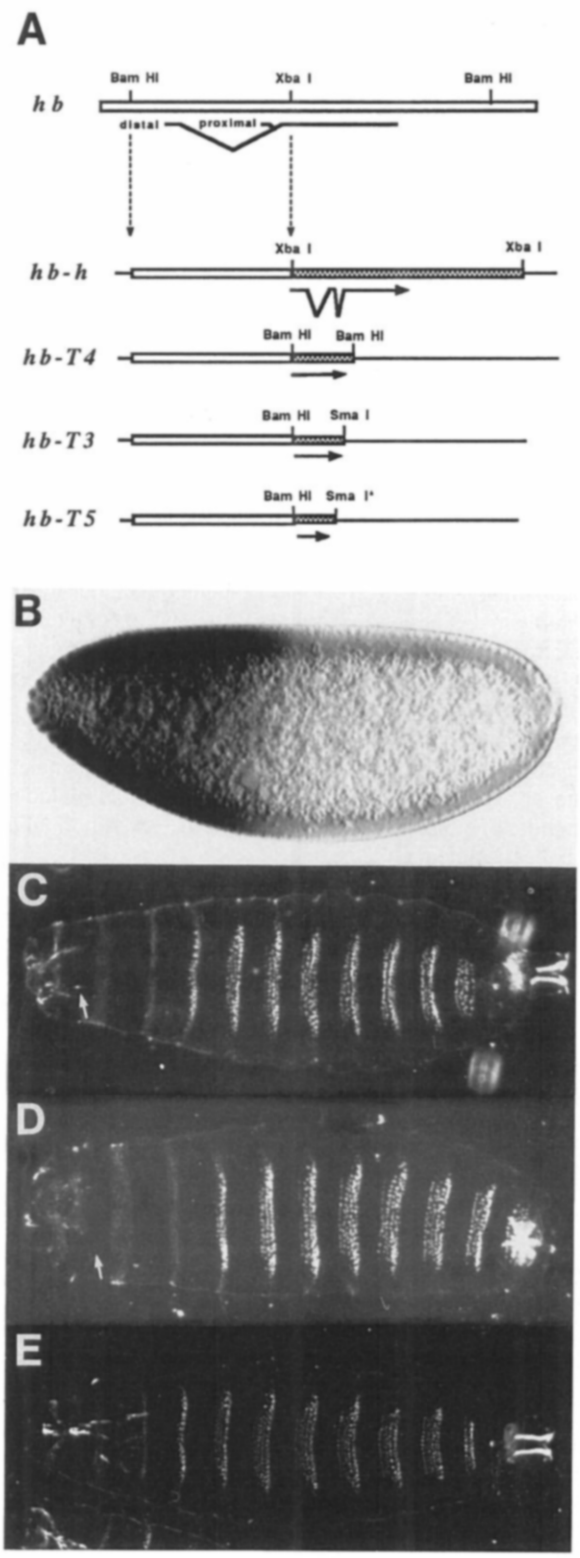

Figure 2. *hb-h* Is Female-Specific Lethal

(A) Restriction maps of the *hb* gene/promoter used for these fusion constructs and the fusions to *h*, *T4*, *T3*, and *T5*. (B) Stage 12 embryo showing ectopic *h* protein expression in the *hb* domain (anterior half of the embryo). (C and D) Cuticular phenotype of female larvae containing one copy of the *hb-h* construct. Ectopic expression of *h* does not lead to gross pattern abnormalities (see also S. M. P. and D. I.-H., submitted), except that the mouthparts are not properly formed (arrows) compared with wild-type larvae (E).

11/12 in the anterior 45% of the embryo (Figure 2B; see also S. M. P. and D. I.-H., submitted). Such embryos show no pattern defects because the ectopic *h* expression is transcriptionally regulated before the end of blastoderm, when pair-rule genes act in defining metameric pattern (see S. M. P. and D. I.-H., submitted for details). (Unless otherwise indicated, *hb-h* embryos/flies refers to embryos/flies heterozygous for a single *hb-h* insert.)

Unexpectedly, we find that only male *hb-h* flies are recovered: a single *hb-h* copy is 99%–100% lethal in females, whereas males are unaffected. Sixteen transformed lines were analyzed, 15 of which give <<1% (0/>1000) transformed females. One transformant line is leaky: female survival is higher than in the other lines but still only 1%. The lethality of female *hb-h* embryos is associated with defective mouthparts that derive from the *hb* domain, i.e., the domain of *h* misexpression (Figures 2C and 2D). Thus, ectopic *h* appears to be affecting a specific sex-dependent process, despite playing no such role during normal development.

The *hb-h* female-specific lethality must be due to premature *h* expression, the *hb* promoter being active two to three cleavage cycles earlier than *h*. Sex determination precedes the establishment of segmental pattern because expression of the X-linked pair-rule gene *runt* is already dosage compensated (Gergen, 1987). *h* must be interfering with an early stage of sex determination that is specifically required in females, e.g., initiation of *Sxl* expression. We therefore tested whether *hb-h* female lethality is due to a failure of *Sxl* expression.

Constitutive *Sxl* Expression Rescues *hb-h* Females

If *hb-h* interferes with initiation of *Sxl* expression, the viability of *hb-h* females should be restored by constitutive *Sxl* expression. We tested this using Sxl^{M1}, a male-lethal allele that causes *Sxl* to be expressed in both sexes, independent of the X:A ratio (Cline, 1978, 1984). Sxl^{M1} females are viable even when derived from *da* mothers, as they still express *Sxl*. In contrast, Sxl^{M1} males are lethal because their inappropriate *Sxl* expression leads to a failure of dosage compensation (e.g., Table 1A).

Sxl^{M1} completely suppresses *hb-h* female lethality, demonstrating that the ectopic *h* interferes with the *Sxl* sex determination switch (Table 1B and C). Indeed, *hb-h*/+; Sxl^{M1}/+ females are fully viable even in the presence of two *hb-h* copies (not shown). Therefore, we analyzed the pattern of *Sxl* expression in *hb-h* embryos in more detail.

Anterior *Sxl* Expression Is Suppressed in Female *hb-h* Embryos

As *hb-h* affects *Sxl* initiation, we expected *Sxl* protein expression to be altered, but only where *h* is ectopically expressed: in the anterior of the embryo. We confirmed this by direct examination of the pattern of *Sxl* protein expression using a female-specific *Sxl* monoclonal antibody against a downstream *Sxl* epitope that is expressed in fe-

Table 1. Constitutive *Sxl* Expression Rescues *hb-h* Female Lethality

Parents		Progeny							
		Females				Males			
♀♀	♂♂	Sxl^{M1}		FM7		Sxl^{M1}		FM7	
A. Control									
Sxl^{M1}/FM7	+/+	53		31		0[a]		13	
B. With G418 selection									
Sxl^{M1}/FM7	*hb-h*#20/+	24		0		0		17	
C. No G418 selection									
		hb-h	Bal.[b]	*hb-h*	Bal.	*hb-h*	Bal.	*hb-h*	Bal.
Sxl^{M1}/FM7	*hb-h*#20/CyO	53	62	0	37	0	0	41	26
Sxl^{M1}/FM7	*hb-h*#31/TM3	46	41	0	23	0	0	33	17

[a] Constitutive *Sxl* expression is lethal in males.
[b] Bal., balancer.

male but not male tissue (D. B., T. Cline, and P. Schedl, submitted). Although *Sxl* is transcribed in both sexes, translation is truncated in males so that only females make full-length active protein (Bell et al., 1988; Salz et al., 1989). Female embryos show uniform staining throughout development, starting from early blastoderm stages (D. B. et al., submitted; see also Figures 3B and 3D). Consistent with previous analysis of *Sxl* transcript structure, *Sxl* protein is not detectable in male embryos (Figures 3A and 3C).

As predicted, female *hb-h* embryos display a unique *Sxl* staining pattern. *Sxl* staining is limited to the posterior of the embryo, a novel pattern representing female *hb-h* embryos in which anterior *Sxl* expression has been suppressed by ectopic *h* (Figure 3E; compare with the complementary pattern of ectopic *h* staining in Figure 2B). *hb-h* females appear never to initiate anterior *Sxl* expression, as posterior-specific staining is first apparent at cleavage cycle 12, the earliest stage at which *Sxl* expression is normally detectable, and is stably maintained thereafter (Figures 3F–3H). Male *hb-h* embryos cannot be distinguished from their wild-type siblings because males never express *Sxl*. *Sxl* expression must become *h* independent before early cleavage stage 14, when normal *h* expression commences.

hb-h Interferes with X Chromosome Counting

Although *h* normally plays no role in sex determination, it is an HLH protein, suggesting that ectopic *h* might inter-

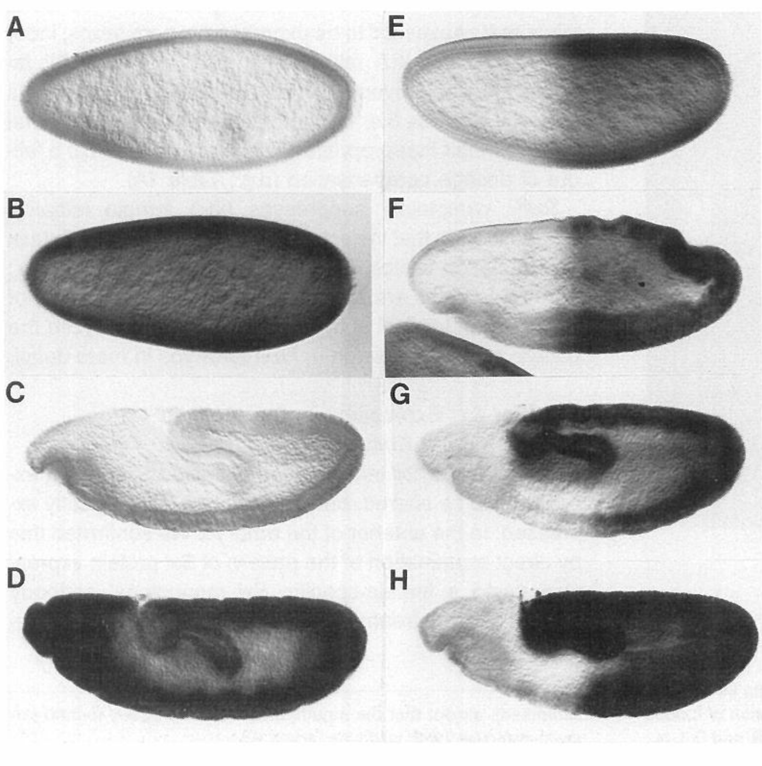

Figure 3. *Sxl* Protein Expression in Wild-Type and *hb-h* Embryos

(A–D) Expression of *Sxl* protein in wild-type male (A and C) and female (B and D) embryos at the cellular blastoderm (A and B) or germband extended (C and D) stages, using a female-specific monoclonal antibody (D. B. et al., submitted). The antibody detects protein in all female cells (B and D) and no male cells (A and C).

(E–H) Expression of *Sxl* in female embryos containing one copy of the *hb-h* construct. The anterior portion of the embryo that is ectopically expressing *h* does not express *Sxl*. This pattern is maintained throughout development: cellular blastoderm (E), early gastrula (F), germband extending 5 hr postfertilization (G), and late germband extension (H). The sex of embryos can be inferred from the staining pattern as described by D. B. et al. (submitted). All embryos were photographed using Nomarski optics and oriented such that anterior is to the left and dorsal is up (also Figure 4).

Table 2. *sis*$^{+}$ but Not *da*$^{+}$ Duplications Rescue *hb-h* Female Lethality

Parents		Progeny			
		Females		Males	
♀♀	♂♂	*hb-h*	CyO	*hb-h*	CyO
A. *sis*$^{+}$ duplications (no G418 selection)					
+/+	*hb-h*#20/CyO	0	53	65	98
Dp sc^{19} (*sis-b*$^{+}$)	*hb-h*#20/CyO	18	34	59	84
In sc^{4L} *sc*^{8R} (*sis-b*$^{+}$)	*hb-h*#20/CyO	49	57	81	113
Dp v^{65b} (*sis-a*$^{+}$)	*hb-h*#20/CyO	41	36	69	76
DD(2)Ha (*sis-a*$^{+}$ + *sis-b*$^{+}$)	*hb-h*#20/CyO	26	19	52	47
B. *da*$^{+}$ or *fs(1)1621*$^{+}$ duplications (with G418 selection[a])					
+/+	*hb-h*#20/CyO	0	–	142	–
p[*da*$^{+}$]/+ [b]	*hb-h*#20/CyO	0	–	146	–
p[*da*$^{+}$]/p[*da*$^{+}$][b]	*hb-h*#20/CyO	0	–	131	–
Dp da^{+}/+ [c]	*hb-h*#20/CyO	0	–	151	–
Dp(2)fs(1)1621	*hb-h*#20/CyO	0	–	193	–

[a] Only *hb-h* containing progeny can survive.
[b] *da*$^{+}$ duplication on P element transposon.
[c] Genetic tandem *da*$^{+}$ duplication.

fere with determination of X:A ratio, a process in which other HLH proteins are implicated. *h* could antagonize general/mandatory factors needed for *Sxl* transcription (e.g., *da*), or it could prevent the action of zygotic chromosome counting elements (*sis-a*, *sis-b*). To distinguish between these alternatives, we asked whether extra wild-type doses of either target gene could suppress *hb-h* female lethality.

Extra numerator elements suppress the female-lethal effects of ectopic *h*, confirming that ectopic *h* interferes with X chromosome counting. *Dp(1;2)v*^{65b} and *Dp(1;2) sc*^{19}, duplications of the X-linked counting elements *sis-a* or *sis-b*, respectively, partially or completely suppress *hb-h* female lethality (Table 2A; Campuzano et al., 1985; Cline, 1988). *In(1)sc*^{4L}*sc*^{8R}, a smaller ~20 kb duplication that also duplicates the *sis-b* gene, is sufficient to allow survival of *hb-h* females (Figure 4A; Table 2A; Cline, 1988; Torres and Sánchez, 1989). Thus, the numerator elements suppress the female-lethal effects of ectopic *h*.

In contrast, maternal *da*$^{+}$ duplications fail to rescue. Table 2B shows that *hb-h* females are still lethal, even if derived from females containing two extra maternal *da*$^{+}$ copies (four copies total; see Experimental Procedures), nor are *hb-h* females rescued by an extra wild-type copy of *fs(1)1621*, a maternal locus implicated in *Sxl* initiation at the level of the autoregulatory feedback loop required for the establishment and proper maintenance of *Sxl* expression (Table 2B; Cline, 1988; Oliver et al., 1988; Steinmann-Zwicky, 1988).

A Model for X Chromosome Counting

Previous evidence regarding the nature of X chromosome counting elements has been somewhat ambiguous. They have usually been regarded as DNA sites that are measured by other *trans*-acting components (Chandra, 1985; Waring and Pollack, 1987; Cline, 1988; see also Hodgkin, 1987a, 1990). However, *sis-b* maps within the AS-C complex, and mutations in the AS-C *T4* protein affect sex determination, suggesting that *sis-b* involves AS-C *T4* protein activity, presumably as a transcriptional regulator (Torres and Sánchez, 1989). Moreover, *wint*, an altered RNA polymerase subunit that reduces transcription of selected genes, interacts with *sis-a*, *sis-b*, and *Sxl*, suggesting that initiation of *Sxl* expression and chromosome counting are transcriptionally regulated (S. M. P. and D. I.-H., submitted).

The effects of *hb-h* are compelling evidence in favor of the latter view and are easily reconciled to the following simple model for the initiation of *Sxl* expression. We propose that *Sxl* initiation depends on its transcriptional activation by HLH protein heterodimers of *da* and *sis* proteins. *da* is contributed maternally and is present in excess. The counting elements (*sis-b*; and predicted for *sis-a*) are transiently expressed in all cells and are quantitatively limiting. As *sis* expression precedes dosage compensation (which can only follow *Sxl* activation), females make twice as much of each "counting protein" as males. Only females would make sufficient *da–sis* heterodimers to activate the *Sxl* autoregulatory feedback loop (see Figure 5, top). Ectopic *h* is female lethal by binding *T4* and other HLH counting proteins, preventing them from binding *da* and activating *Sxl* transcription (see Figure 5, middle). This model leads to several testable predictions: First, if *sis-b* is (or includes) the AS-C *T4* protein, its ectopic expression should cause inappropriate *Sxl* activation and thereby male lethality. Ectopic expression of other HLH AS-C proteins contributing to X chromosome counting would have similar consequences. Second, ectopic *Sxl* expression will be restricted to domains of ectopic *T4* expression. Third, if *T4* acts through a heterodimer with *da*, male lethality will be dependent on maternal *da*.

***hb-T4* Causes Male Lethality**

In the rest of this paper, we describe the effects of ectopic AS-C *T3*, *T4*, and *T5* expression on sex determination. Using an analogous approach to that described above for *h*, we constructed *hb-T3*, *hb-T4*, and *hb-T5* fusion genes, in which the *hb* promoter drives ectopic expression of individual AS-C transcripts in the anterior half of the embryo (see Figure 2A). Several transformed lines were recovered from each construct and examined for sex-specific effects on viability (Table 3; see Experimental Procedures).

Our major prediction is met: *hb-T4* leads to male lethality. Males with two *hb-T4* copies are unconditionally lethal, whether from the same or different insertions (Table 3A and D). For males transformed with one copy of *hb-T4*, the penetrance is incomplete and variable between individual lines (Table 3A). Female viability is unaffected, even by two *hb-T4* copies, indicating that the *hb-T4* lethality is male specific. The low proportion (6%) of homozygous males recovered in one line (*hb-T4*#21) presumably reflects reduced *T4* expression associated with this specific insertion event.

hb-T4 may be less potent when integrated into the X chromosome, causing no marked reduction in male viability when present in one (hemizygous) copy (Table 3A). This may be due to reduced *T4* expression as, for two of the three X-linked lines, males containing two *hb-T4* copies

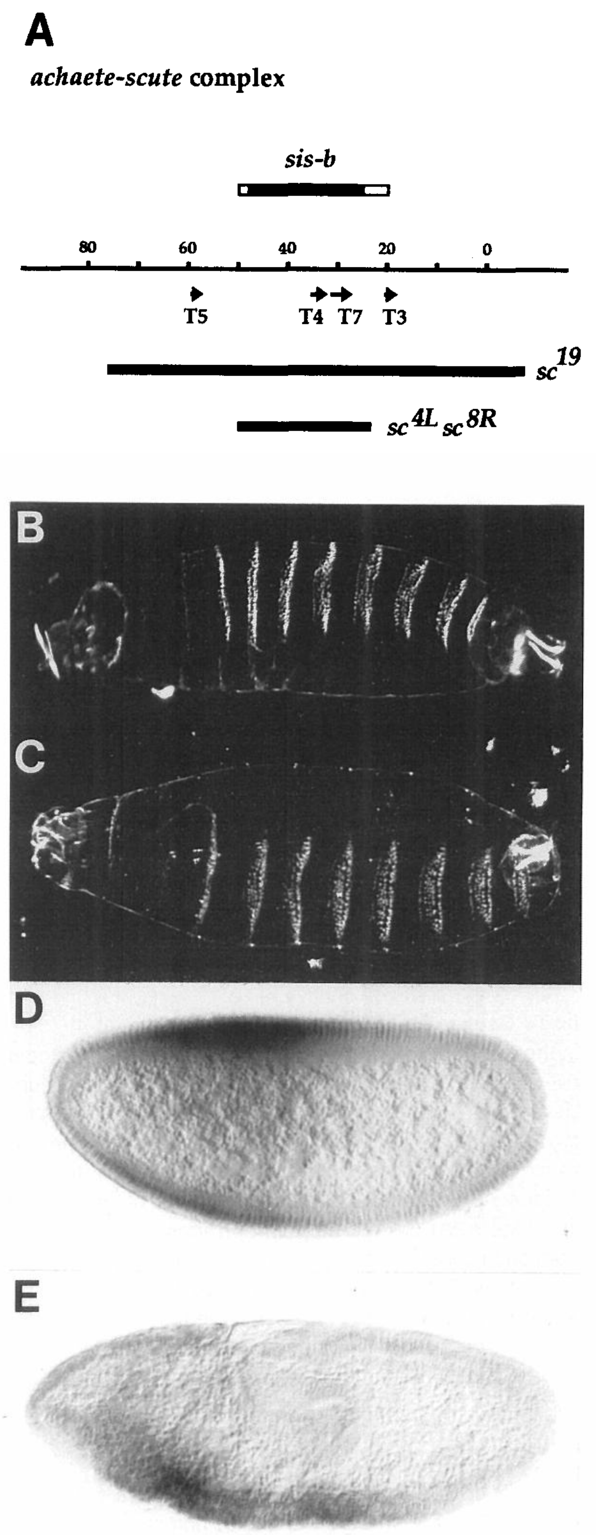

Figure 4. *hb-T4* Is Male-Specific Lethal and Causes Ectopic *Sxl* Expression

(A) Molecular map of AS-C showing the positions of the *T3*, *T4*, *T5*, and *T7* transcripts and the genetic delimitation of the *sis-b* locus (Campuzano et al., 1985; Villares and Cabrera, 1987; Alonso and Cabrera, 1988; Cline, 1988). The extent of the *sis-b* duplications (*Dp(2)sc*^{19} and *In(1)sc*^{4L}*sc*^{8R}) and the genetically defined *sis-b* function (Campuzano et al., 1985; Cline, 1988) are included.

(B and C) Cuticle phenotype of male embryos containing one copy of the *hb-T4* construct. The thoracic and abdominal pattern is unaffected; however, the mouthparts are not properly formed (see Figure 2E).

(D and E) Male embryos containing one copy of the *hb-T4* construct stained with anti-*Sxl* antibodies. Ectopic *Sxl* expression is seen in the anterior half of the embryo where *T4* is ectopically expressed. Cellular blastoderm stage is shown in (D), and the germband extended stage in (E). The ectopically expressed *Sxl* protein is not uniformly expressed in the *hb* domain nor is the level of expression as high as in wild-type females.

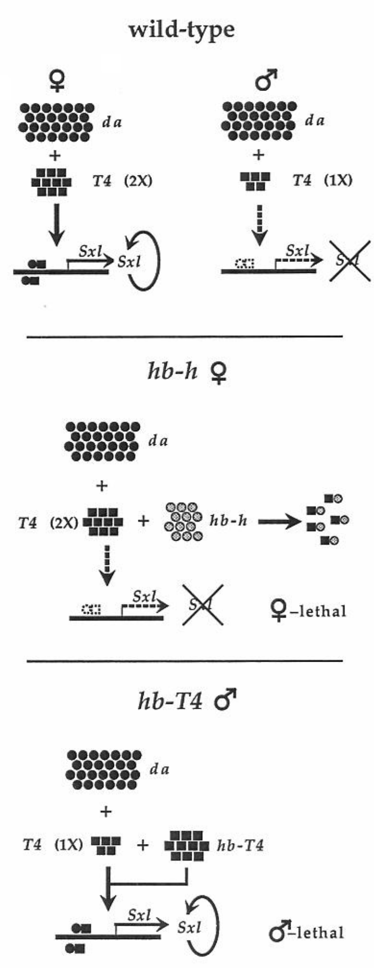

Figure 5. Models of Sex Determination, Illustrating Sex Determination under the Control of Competing HLH Proteins

(Quantitative details are only intended illustratively; male/female differences would be amplified by sequestering of active *sis* protein by the denominator elements [see text].)

(Top) Under normal wild-type conditions, the *da* protein is maternally deposited and present in excess. Twice as much *T4* protein is made in females with two X chromosomes compared with males that have one X chromosome. Males make insufficient *da–T4* heterodimers to initiate the *Sxl* autoregulatory feedback loop and stable *Sxl* expression.

(Middle) *hb-h* reduces the levels of active *T4* in wild-type females by forming nonfunctional heterodimers and preventing the establishment of stable *Sxl* expression in the domains where *h* is ectopically expressed.

(Bottom) When extra *T4* protein is provided by the *hb-T4* construct, males can now make sufficient *da–T4* heterodimers to establish *Sxl* expression stably.

Table 3. *hb-T4*, but Not *hb-T3* and *hb-T5*, Is Homozygous Male Lethal

		Progeny			
		Females		Males	
		(No. of *hb-T4* Copies)			
Transformant Line		1	2	1	2
A. *hb-T4*					
#20/CyO		32	11	24	0
#21/CyO		54	32	44	2[a]
#22/CyO		46	7	37	0
#23/CyO		35	26	15	0
#24/CyO		159	184	23	0
#30/TM3		6	0[b]	1	0
#31/TM3		27	0[b]	2	0
#32/TM3		19	16	4	0
#33/TM3		26	21	9	0
#34/TM3		144	123	12	0
#10[c]		–	36	16	–
#11[c]		–	22	16	–
#15[c]		–	41	28	–
B. *hb-T3*[d]					
#20		–	64	–	41
#30		–	47	–	28
C. *hb-T5*[e]					
#20		–	53	–	53
#30		–	28	–	36
D. Transheterozygous *hb-T4* lines					
♀♀	♂♂				
#24/#24	#34/TM3	47	39	7	0
#23/#23	#24/CyO	54	52	9	0
#10/#10	#24/CyO	39	43	22	5
#11/#11	#24/CyO	52	36	27	0
#15/#15	#24/CyO	60	69	29	5

[a] "Leaky."
[b] *hb-T4*#30 and *hb-T4*#31 are homozygous lethal insertion events.
[c] Transformant lines with X chromosome insertions.
[d] Only 2 *hb-T3* lines of 9 are shown.
[e] Only 2 *hb-T5* lines of 9 are shown.

are somewhat viable if one copy is X linked (Table 3D). We cannot currently explain why X linkage should affect *hb-T4* activity, although it may reflect an unusual transcription environment associated with X chromosomes before they become dosage compensated.

Flies containing either *hb-T3* or *hb-T5* display no significant effects on viability. Males and females transformed with either construct are equally viable, whether heterozygous or homozygous for a fusion gene (Table 3B and C), suggesting that AS-C *T4* is the major sex determinant within AS-C. We are currently testing whether AS-C *T3* and/or AS-C *T5* plays a subsidiary role as a weak counting element.

As expected, posterior cuticle pattern is normal in *hb-T4* embryos, but head morphology is defective, presumably owing to improper dosage compensation (see Figures 4B and 4C). This is consistent with male lethality being due to anterior AS-C *T4* misexpression.

Ectopic AS-C *T4* Induces *Sxl* Expression

If *hb-T4* acts by inducing *Sxl* expression, the male lethality should be suppressed by mutations in *Sxl*. This is confirmed in appropriate crosses showing that *hb-T4; Sxl* male embryos are fully viable (Table 4A). Thus, AS-C *T4* acts directly on *Sxl* expression rather than on downstream genes.

We also visualized the *Sxl* misexpression by staining *hb-T4* embryos directly for full-length *Sxl* protein; 21% (63/297) of embryos show *Sxl* staining that is restricted to the anterior half, representing a novel class of male embryos (see Figures 4D and 4E). This ectopic expression would prevent dosage compensation and is presumably the cause of the male lethality.

Consistent with the partial viability of (heterozygous) *hb-T4* males, *Sxl* induction by *hb-T4* is incomplete. *Sxl* expression in *hb-T4* males is less than in wild-type female embryos, and not all anterior cells express ectopic *Sxl* (compare Figures 4D and 4E with Figure 3). Surprisingly, intermediate *Sxl* levels are stably maintained, suggesting that other mechanisms can modulate levels of *Sxl* expression, as *Sxl* autoregulation would predict only two stable outcomes, Sxl^{ON} or Sxl^{OFF}.

da and *T4* Are Both Required for Induction of *Sxl* Expression

In vitro experiments have suggested that the HLH transcription factors may act primarily as protein–protein heterodimers. In particular, the E12 protein, a vertebrate homolog of *da*, binds the κE2 enhancer binding site much more efficiently as a heterodimer with an assortment of other HLH proteins, including *MyoD* and AS-C *T3* (Murre et al., 1989a, 1989b). This suggests that *da–sis* heterodimers may be the active species in activating *Sxl* expression and that *hb-T4* activation of *Sxl* should be dependent on maternal *da* protein. Alternatively, maternal *da* could be required for zygotic *sis* expression, in which case *hb-T4* would act independently of *da*.

By analyzing *hb-T4* progeny from *da* mothers, we showed that *da* and AS-C *T4* are both required for *Sxl* induction. Table 4B shows that *hb-T4* sons are fully viable from da^1 mothers, whereas *hb-T4* males from control mothers are only 9%–16% viable. The *da* requirement is indeed maternal, as the reverse cross (using *da* fathers) shows no male rescue (Table 4B).

Surprisingly, heterozygosity for *da* also suppresses *hb-T4* male lethality (Table 4B), although one da^+ dose suffices for normal *Sxl* activation. We presume that this sensitivity to *da* dosage is a consequence of the inefficient induction of *Sxl* by *hb-T4*. More detailed analysis of *Sxl* expression patterns will indicate the extent to which reducing *da* levels influences the efficiency of *Sxl* induction.

In Vivo Competition between *T4* and *h* Allows Survival of *hb-h* Females

We have suggested that *hb-h* acts by sequestering the limited levels of *sis*, preventing the formation of *da–sis* heterodimers. This predicts that *T4* overexpression should overcome the *hb-h* female lethality. We demonstrated such competition in vivo by ectopically expressing both *h* and *T4* in *hb-h*; *hb-T4* embryos. Table 5 shows that *hb-h*; *hb-T4* females are fully viable, indicating that the ectopic *T4* successfully competes with the ectopic *h*.

Table 4. *hb-T4* Male Lethality Is Suppressed by Mutations in *Sxl* or *da*

A. *hb-T4* Male Lethality is Rescued by *Df Sxl*

		Male Progeny			
	Female Progeny	*Df Sxl*		Bins or FM7	
Cross	(Total)	*hb-T4*	Balancer	*hb-T4*	Balancer
Sxl^{f1}/Bins × *hb-T4#24*/CyO	64	42	23	3	26
Sxl^{f1}/Bins × *hb-T4#34*/TM3	67	31	12	5	11
Sxl^{7B0}/FM7 × *hb-T4#24*/CyO	29	19	22	1	4
Sxl^{7B0}/FM7 × *hb-T4#34*/TM3	46	26	37	4	8

B. Maternal but Not Paternal Loss of da^+ Rescues *hb-T4* Male Lethality

Parents		Progeny				%
		Females		Males		Male
♀♀	♂♂	*hb-T4*	Balancer	*hb-T4*	Balancer	Viability
Maternally depleted *da*						
+/+	#24/CyO	143	139	26	163	16
da^2/+	#24/CyO	214	196	184	136	~100
da^1/+	#24/CyO	266	112	238	105	~100
da^1/da^1	#24/CyO	0	0	171	162	~100
+/+	#34/TM3	154	89	12	137	9
da^2/+	#34/TM3	83	111	187	138	~100
da^1/+	#34/TM3	198	54	144	151	~100
da^1/da^1	#34/TM3	0	0	168	78	~100
Paternally depleted *da*						
		da da^+	*da* da^+	*da* da^+	*da* da^+	*da* da^+
#24/CyO	+/+	82	72	21	73	29
#24/CyO	da^1/Sco	57 44	39 37	3 14	61 53	5 26
#24/CyO	da^2/Sco	39 41	44 48	9 5	54 36	17 14
#34/TM3	+/+	79	66	17	54	31
#34/TM3	da^1/Sco	63 52	49 36	11 8	47 59	23 14
#34/TM3	da^2/Sco	31 36	33 33	6 6	38 42	16 14

hb-h appears less effective in suppressing the male lethality of *hb-T4*, failing to enhance the viability of hemizygous *hb-T4#24* males and only partially restoring the viability of *hb-T4#34* males (Table 5). Preliminary experiments indicate that *hb-h* suppresses the male lethality of two *hb-T4* copies (not shown). Nevertheless, *hb-h* appears to compete more successfully with endogenous *T4* than with *hb-T4*. Further analysis is required to explain these results, but they may indicate that *hb-T4* expresses more *T4* at blastoderm than does the endogenous *T4* gene, which is active only briefly during blastoderm (Romani et al., 1987).

Discussion

We have shown that Drosophila sex determination and, in particular, X chromosome counting is sensitive to mis-

Table 5. *hb-h* and *hb-T4* Rescue Their Reciprocal Lethalities

	Progeny			
	Females		Males	
Cross	*hb-h*	CyO	*hb-h*	CyO
+/+ ♀♀ × *hb-h#20*/CyO ♂♂	0	37	46	41
hb-T4#24/#24 ♀♀ × *hb-h#20*/CyO ♂♂	61	65	23	21
hb-T4#34/#34 ♀♀ × *hb-h#20*/CyO ♂♂	53	53	35	15

regulated expression of HLH transcription factors. Ectopic *h* expression under the control of the *hb* promoter kills female embryos because the premature *h* protein prevents *sis* counting elements from initiating *Sxl* expression (Figure 5, middle). Note that *h* plays no normal role in sex determination, illustrating the potential dangers of deducing wild-type function from the effects of genetic misexpression. In complementary experiments, ectopic AS-C *T4* expression kills males by raising the effective X:A ratio and inappropriately activating *Sxl* (Figure 5, bottom).

Previous results had suggested that AS-C *T4* is a major contributor to the *sis-b* numerator element whose dosage determines the state of *Sxl* expression (Cline, 1988; Torres and Sánchez, 1989). This is confirmed and extended by our results showing that ectopic *T4* expression can directly activate *Sxl*. *hb-T4* only activates *Sxl* in the *hb* transcriptional domain, showing that expression is crucial for *T4* action, i.e., *sis-b* is not merely acting as a DNA site (Chandra, 1985; Waring and Pollack, 1987; Cline, 1988). Rather, it encodes an HLH protein whose role it is to activate *Sxl* transcription.

Our model for the early events that initiate sex determination is as follows (Figure 5, top): Heterodimers between *da* and counting proteins (such as *sis-a* and *sis-b*) promote transcription of an early, initiator *Sxl* transcript that encodes active *Sxl* protein. The levels of *sis-a*, *sis-b*, etc. are limiting, determining the efficiency of early *Sxl* transcription and thereby the amounts of initiator protein. Only females, with two copies of the X-linked counting genes, encode sufficient *Sxl* protein to allow stable activation of female-specific autocatalytic splicing.

Sex Determination Occurs during the Early Blastoderm Stage

Normal *h* and AS-C expression does not cause sex-specific lethality. *h* expression during the late blastoderm stage does not repress *Sxl*, nor does *T4* expression at gastrulation induce *Sxl*. Thus, sex determination is irreversibly determined before nuclear cycle 14, consistent with previous results showing that dosage compensation is established before pair-rule segmentation genes are active (Gergen, 1987).

Sex must be determined between blastoderm stages 10 and 14 when *Sxl* expression is sensitive to *hb*-encoded ectopic HLH expression. This is also consistent with the onset of detectable staining for *Sxl* protein at stage 12 (D. B. et al., submitted) and the presence of homogeneously distributed AS-C *T4* transcripts during the early cleavage stages (Cabrera et al., 1987; Romani et al., 1987).

HLH Proteins Initiate *Sxl* Expression

The effects of inappropriate HLH protein expression indicate that *Sxl* initiation is transcriptionally regulated (see also Keyes et al., submitted). HLH proteins are nuclear transcription factors, as has been most directly demonstrated for the murine proteins E12, E47, and *MyoD* (Davis et al., 1987; Murre et al., 1989a, 1989b). Constitutive *Sxl* alleles do not require maternal *da*; indeed they suppress maternal *da* mutations, so expression of the late *Sxl* transcript is likely to be independent of the HLH transcription factors needed for initiation (Cline, 1978, 1984). Rather, AS-C *T4*'s primary target must be a specific early promoter that drives initiator *Sxl* activity (Bell et al., 1988; Salz et al., 1989; Keyes et al., submitted).

The inhibition of *Sxl* expression by ectopic *h* reinforces the above interpretation. Duplication of either *sis-a*$^+$ or *sis-b*$^+$ restores *hb-h* female viability, indicating that *h* interferes with assessing X:A ratio. Duplications of *da*$^+$ do not rescue *hb-h* females, nor are *da* levels normally limiting for sex determination; thus, *h* appears not to be inhibiting *da*. Rather, *hb-h* prevents counting proteins from activating *Sxl*. From these results, together with previous evidence for the partial interchangeability of *sis-a* and *sis-b* (Cline, 1988), we would expect that other (probably all) counting elements are also HLH proteins.

Ectopic *T4* is not fully effective at inducing *Sxl* expression. *hb-T4* males show intermediate levels of *Sxl* expression and are only partially lethal. This is unlikely to reflect limited AS-C *T4* expression, as *hb* is efficiently expressed from the onset of the syncytial blastoderm stage (see S. M. P. and D. I.-H., submitted). *Sxl* is X linked, so wild-type females will make more *Sxl* protein than *hb-T4* males because they have twice the *Sxl* gene dose. In addition, full *Sxl* transcription may require AS-C *T4* to cooperate with other (HLH) counting elements that are not misexpressed in *hb-T4* males.

Previous experiments have suggested that other AS-C genes might contribute to *sis* activity (Cline, 1988; Torres and Sánchez, 1989). Ectopic expression of two other AS-C HLH proteins, *T3* and *T5*, is not associated with sex-specific lethality, suggesting that they are not major activators or inhibitors of *Sxl* expression. However, AS-C *T4* and *T5* appear somewhat interchangeable in bristle patterning (Balcells et al., 1988), implying that they can share functional roles. We are currently analyzing *hb-T3* and *hb-T5* transformants in more detail to determine whether they play minor roles in influencing chromosome counting.

How is a 2-fold difference in *sis* dosage converted into bistable *Sxl* expression? We do not yet know how the threshold is set, but only females make sufficient *Sxl* protein to achieve autoregulation. The limited time in which *sis* proteins can act argues that they function together rather than in sequence. We anticipate that synergistic interactions between *sis* proteins, e.g., through multiple binding sites, will amplify differences in initiator *Sxl* transcript levels between females and males, augmented by dosage differences in *Sxl* itself.

Indeed, effective *sis* concentrations will differ more than 2-fold owing to titration by autosomal denominator elements. If *hb-h* is a model for how denominator genes act, then they will encode autosomal HLH proteins that bind to and sequester *sis* proteins (e.g., see Figure 5 [middle], though *h* itself is expressed too late to be a denominator gene). Alternatively, denominator elements may be DNA binding sites, not protein-coding genes. In this case, the limiting amounts of *da–sis* dimers would partition between the *Sxl* gene and alternative binding sites dispersed through the rest of the genome, the latter providing a good measure of autosomal gene content.

In Vivo Evidence for HLH Heterodimers (In Vivo Competition between HLH Proteins)

We propose that *da–sis* heterodimers are the active components in chromosome counting. Thus, *hb-T4*'s male lethality is dependent on *da*, and the *hb-h* female lethality is explained by ectopic *h* protein disrupting *da–sis* heterodimerization. Further evidence comes from in vitro heterodimerization and DNA binding of *da* and AS-C *T3* and from the genetic interactions between *da* and AS-C (Moscoso del Prado and Garcia-Bellido, 1984a, 1984b; Murre et al., 1989a, 1989b).

h may function similarly in patterning adult sensory organs, where it prevents the development of ectopic bristles (Falk, 1963; Botas et al., 1982; Moscoso del Prado and Garcia-Bellido, 1984a, 1984b). Although the exact basis of sensory cell selection is not yet clear, it appears to be dependent on the balance between different HLH proteins. AS-C "proneural" genes promote sensory organ development, whereas *h* and *extramacrochaetae* (*emc*) expression represses bristle formation (for reviews see Dambly-Chaudière et al., 1988; Garcia-Alonso and Garcia-Bellido, 1988; Ellis et al., 1990; Garrell and Modolell, 1990). We believe that *h* is likely to act by antagonizing the formation of *T5*–HLH heterodimers, consistent with genetic interactions between *h* and *achaete* (*ac*) (*T5*) (Botas et al., 1982). Earlier suggestions that *h* is a regulator of *ac* expression appear unlikely because *h* mutations that cause development of ectopic bristles do not alter the patterns of *ac* expression (Romani et al., 1989). Rather, bristle cells in the wild-type wing-disc cease *h* expression, as if *h* expression is inconsistent with sensory organ fate. *h* is the only well-characterized segmentation gene not expressed in the developing embryonic nervous system, perhaps to ensure that it does not interfere with HLH-mediated disposition of neural cell fates (Carroll et al., 1988; Carroll and Whyte, 1989; Hooper et al., 1989).

As indicated above, *h* acts as a negative transcriptional regulator, presumably because it forms transcriptionally inactive heterodimers. Their lack of activity could be due to an inability to bind DNA or to their failure to activate transcription. DNA binding activity in HLH proteins and other heterodimeric transcriptional activators (e.g., *jun–fos*) appears to depend on conserved basic domains that lie adjacent to potential heterodimerization helices (Halazonetis et al., 1988; Kouzarides and Ziff, 1988; Nakabeppu et al., 1988; Tapscott et al., 1988; Benezra et al., 1990; Davis et al., 1990). This domain is lacking in two other HLH repressors, *emc* and Id, leading to suggestions that they may repress by forming heterodimers that do not bind DNA (Benezra et al., 1990; Ellis et al., 1990; Garrell and Modolell, 1990). *h* may represent a different class of HLH repressor as it retains a basic region such that *h* heterodimers retain DNA binding but lack transcriptional activation. We are currently analyzing mutant *h* proteins to define its functionally important structural domains.

HLH Proteins and Segmentation

The genetic evidence demonstrates that *h* cannot function in sex determination. Nevertheless, the unexpected interference with chromosome counting helps us consider *h*'s role in regulating segmentation gene expression. The same *h* protein that hinders *Sxl* activation is active in segmentation and bristle patterning (Rushlow et al., 1989; S. M. Pinchin, personal communication), arguing for similar regulatory mechanisms in all three situations. In the last two cases, *h* acts by sequestering activator HLH transcription factors: *sis*, in the case of sex determination, and (presumably) *T5* in bristle patterning.

We suggest that *h* plays a similar role in segmentation: binding and inhibiting an HLH protein that otherwise activates *ftz* transcription. Such a factor is likely to be ubiquitously distributed within the embryo because *h* already conveys the spatial information for striping *ftz*. No candidate genes for *h*'s "partner" have yet been identified: it cannot be an AS-C protein, as they are not required for segmentation (Cabrera et al., 1987; Romani et al., 1987, 1989), nor does *da* have maternal or zygotic effects on segmentation (Cronmiller and Cline, 1987; Cronmiller et al., 1988). Genetic screens for further zygotic segmentation genes have not uncovered novel *ftz*-regulating genes (Vavra and Carroll, 1989), suggesting that the partner protein may be maternally deposited in the egg. If it also regulates zygotic transcription, mutant females will be lethal and its maternal effects on segmentation would be masked. In this event, alternative strategies for the identification of *h*'s partner protein will be required.

Chromosome Counting in Nematodes and Mammals

Our results may also be relevant to chromosome counting in other organisms that determine sex by X:A ratio (see Bull, 1983). We have excluded Drosophila counting elements being nontranscribed DNA "sites." A similar controversy has arisen over how X chromosomes in C. elegans are counted. We would favor the action of transcribed elements (Hodgkin, 1987a, 1990) rather than reiterated X-linked DNA sequences (McCoubrey et al., 1988).

Mammals also measure X:A ratio, not to define sex, but to regulate dosage compensation. Only a single X chromosome remains transcriptionally active in mice and humans, even in diploid animals with extra X chromosomes (for reviews see Lyon, 1972; Lucchesi, 1978). This does not reflect an absolute requirement for one active X chromosome, but, like Drosophila, serves to preserve the balance of gene expression. Thus, tetraploid human embryos retain two active X chromosomes, indicating that their num-

Table 6. Fly Stocks

Stock	Provider	Reference
Sxl^{M1}/FM7	R. Nöthiger	Cline (1978)
Df(1)sc^{19}, f^{36a}/CDX1; Dp(2)sc^{19}, b pr/SM5	C. Cabrera	Lindsley and Zimm (1987)
In(1)sc^{4L}sc^{8R},y sc^{4}sc^{8}	Bowling Green	Lindsley and Zimm (1987)
Dp(1;2)v^{65b},v^{m} sis-a^{+} m^{+}	T. Cline	Cline (1988)
DD(2)Ha, Dp sis-a^{+} Dp sis-b^{+}	T. Cline	Cline (1988)
y cm Df(1)Sxl^{7B0}/FM7	T. Cline	Salz et al. (1987)
cm Sxl^{f1} ct/Bins	T. Cline	Cline (1978)
Dp(M^{+},da^{+})18/Df(2L)J-der-39, M	C. Cronmiller	Cronmiller and Cline (1986)
p[*w^{+},da^{+}*]/*CyO*, p[*w^{+},da^{+}*]	H. Vässin[a]	Personal communication
T(1,2)w^{84b13}, Dp(fs1621)$^{+}$	T. Cline	Lindsley and Zimm (1987)
cl da^{1} cn bw/CyO	C. Cronmiller	Lindsley and Zimm (1985)
y; cl da^{2}/CyO	C. Cronmiller	Cronmiller and Cline (1987)
Dp(3;Y;X)M2, mwh^{+} emc^{+} re^{+},y f^{36a}/FM6	L. Garcia-Alonso	Garcia-Alonso and Garcia-Bellido (1988)
mwh emc^{FX119} (jv)/TM2	L. Garcia-Alonso	Garcia-Alonso and Garcia-Bellido (1988)

Flies were cultured on yeast, maizemeal, molasses, malt extract, and agar medium, at 25°C unless otherwise stated. All crosses shown were done at 18°C and 25°C with similar results. The data shown are for 25°C unless otherwise stated.
[a] The p[*w^{+},da^{+}*] stocks are P element transformants containing sufficient *da* sequences to rescue the *da^{-}* phenotype.

ber depends on X:A ratio (Carr, 1971) and raising the possibility that chromosome counting strategies are similar in mammals and Drosophila.

Experimental Procedures

Fly Stocks
See Table 6 for details.

Constructs
The 4.7 kb BamHI–XbaI fragment containing the *hb* promoter and all but 10 bases of the 5′ untranslated leader sequences was subcloned into the blunt-ended SalI site of *pUChsneo* (Steller and Pirrotta, 1985). This vector, *hbneo*, drives anterior zygotic expression of coding sequences inserted at unique BamHI or SmaI polylinker sites adjacent to the *hb* promoter (S. M. P. and D. I.-H., submitted). For *hb-h*, the 6.5 kb XbaI genomic fragment, including 230 bases of 5′ untranslated leader sequences (Rushlow et al., 1989), was subcloned into the blunt-ended BamHI site of *hbneo*. The genomic fragments containing the AS-C genes were obtained from cloned genomic DNA (gift of C. Cabrera) by polymerase chain reaction amplification using oligonucleotide primers to introduce appropriate restriction sites at their 5′ termini (Erlich, 1989). The amplified fragments were as follows: *hb-T3*, −85 bp to +1380 bp subcloned into the BamHI–SmaI sites of *hbneo*; *hb-T4*, −125 bp to +1485 bp with BamHI termini; *hb-T5*, −65 bp to +1050 bp subcloned into the BamHI–SmaI sites of *hbneo*. The appropriate orientation for all clones was determined by restriction enzyme analysis.

Germline Transformation
bw; st embryos were transformed by injection with a mixture of recombinant plasmid (500 μg/ml) and helper plasmid (100 μg/ml), as described by Spradling (1986). The *bw; st* G_0 adults were outcrossed to wild type and selected on standard medium supplemented with Geneticin G418 Sulphate (GIBCO, 1.5 mg/ml but varied according to batch). G_1 *bw/+; st/+* progeny were mapped by back-crossing crossed to *bw; st* on G418 food. This retested their drug resistance and assigned the insert to a specific chromosome, allowing construction of homozygous or balanced stocks. All *neo*-resistant transformants were confirmed by polymerase chain reaction (Erlich, 1989) using primers specific to the *neo* portion of the P element transformation vector.

Embryo Analysis
Embryos were prepared and analyzed as described by Wieschaus and Nüsslein-Volhard (1986). Immunohistochemical detection of *h* and *Sxl* was performed essentially as described by Macdonald and Struhl (1986), using biotinylated secondary antibodies and avidin–biotin–horseradish peroxidase complexes (Vector Laboratories, Inc.). The rabbit anti-*h* polyclonal antibodies used in this study were generously provided by K. Hooper (Hooper et al., 1989). The mouse anti-*Sxl* monoclonal antibodies will be described elsewhere (D. B. et al., submitted). All secondary antibodies were obtained from Jackson ImmunoResearch Labs (West Grove, PA). The stained embryos were dehydrated in 100% ETOH and mounted under a coverslip in methacrylate mounting medium (JB-4, Polysciences) that was polymerized under CO_2 for 1–2 hr at room temperature.

Acknowledgments

We should like to thank Rolf Nöthiger for suggesting the Sxl^{M1} experiment; Tom Cline, Jim Erickson, Kees Murre, Suki Parks, and Paul Schedl for sharing unpublished information, suggestions, and many helpful discussions; and Juan Modolell and Lucas Sánchez for sharing unpublished information. We should also like to thank Tom Cline, Carlos Cabrera, Mike Caudy, Claire Cronmiller, Luis Garcia-Alonso, Kate Hooper, Juan Modolell, Diethard Tautz, Harold Vässin, and the Bowling Green Stock Center for fly stocks, antibodies, and DNA clones used in this study, and Iain Goldsmith for oligonucleotide synthesis. We are very grateful to Karen Downs, Peter Gergen, Howard Lipshitz, Rolf Nöthiger, Suki Parks, and Detlef Weigel for critical reading and comments on the manuscript. We thank our colleagues in the lab (Kate Hooper, Sheena Pinchin, Mark Wainwright, Michael Lardelli, Guy Riddihough, and Ilen Davis), in the ICRF Developmental Biology Unit, at Caltech, and in the fly groups at Princeton for suggestions and stimulating discussions during the course of this work. D. B. was supported by an EMBO postdoctoral fellowship. S. M. P. was supported by a postdoctoral fellowship from the Helen Hay Whitney Foundation.

The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked "*advertisement*" in accordance with 18 USC Section 1734 solely to indicate this fact.

Received June 4, 1990; revised September 3, 1990.

References

Alonso, M. C., and Cabrera, C. V. (1988). The *achaete-scute* gene complex of *Drosophila melanogaster* comprises four homologous genes. EMBO J. *7*, 2585–2591.

Baker, B. S., and Belote, J. M. (1983). Sex determination and dosage compensation in *Drosophila melanogaster.* Annu. Rev. Genet. *17*, 345–393.

Baker, B. S. (1989). Sex in flies: the splice of life. Nature *340*, 521–524.

Balcells, L., Modolell, J., and Ruíz-Gómez, M. (1988). A unitary basis for different *Hairy-wing* mutations of *Drosophila melanogaster.* EMBO J. *7*, 3899–3906.

Bell, L. R., Maine, E. M., Schedl, P., and Cline, T. W. (1988). *Sex-lethal*, a Drosophila sex determination switch gene, exhibits sex-specific RNA splicing and sequence similarity to RNA binding proteins. Cell *55*, 1037–1046.

Benezra, R., Davis, R. L., Lockshon, D., Turner, D. L., and Weintraub, H. (1990). The protein Id: a negative regulator of helix-loop-helix DNA binding proteins. Cell *61*, 49–59.

Boggs, R. T., Gregor, P., Idriss, S., Belote, J. M., and McKeown, M. (1987). Regulation of sexual differentiation in D. melanogaster via alternative splicing of RNA from the *transformer* gene. Cell *50*, 739–747.

Botas, J., Moscoso del Prado, J., and Garcia-Bellido, A. (1982). Gene dose titration analysis in the search for trans-regulatory genes in *Drosophila*. EMBO J. *1*, 307–310.

Bridges, C. B. (1921). Triploid intersexes in *Drosophila melanogaster.* Science *54*, 252–254.

Bridges, C. B. (1925). Sex in relation to chromosomes and genes. Am. Nat. *59*, 127–137.

Bull, J. J. (1983). Evolution of Sex Determining Mechanisms (Menlo Park, California: Benjamin/Cummings).

Cabrera, C. V., Martinez-Arias, A., and Bate, M. (1987). The expression of three members of the *achaete-scute* gene complex correlates with neuroblast segregation in Drosophila. Cell *50*, 425–433.

Campuzano, S., Carramolino, L., Cabrera, C. V., Ruíz-Gómez, M., Villares, R., Boronat, A., and Modolell, J. (1985). Molecular genetics of the *achaete–scute* gene complex of D. melanogaster. Cell *40*, 327–338.

Carr, D. H. (1971). Chromosome studies in selected spontaneous abortion: polyploidy in man. J. Med. Genet. *8*, 164–174.

Carroll, S. B., and Scott, M. P. (1986). Zygotically active genes that affect the spatial expression of the *fushi tarazu* segmentation gene during early Drosophila embryogenesis. Cell *45*, 113–126.

Carroll, S. B., Laughon, A., and Thalley, B. S. (1988). Expression, function and regulation of the *hairy* segmentation protein in the *Drosophila* embryo. Genes Dev. *2*, 883–890.

Carroll, S. B., and Whyte, J. S. (1989). The role of the *hairy* gene during *Drosophila* morphogenesis: stripes in imaginal discs. Genes Dev. *3*, 905–916.

Caudy, M., Grell, E. H., Dambly-Chaudière, C., Ghysen, A., Jan, L. Y., and Jan, Y. N. (1988a). The maternal sex determination gene *daughterless* has zygotic activity necessary for the formation of peripheral neurons in *Drosophila*. Genes Dev. *2*, 843–852.

Caudy, M., Vässin, H., Brand, M., Tuma, R., Jan, L. Y., and Jan, Y. N. (1988b). *daughterless*, a Drosophila gene essential for both neurogenesis and sex determination, has sequence similarities to *myc* and the *achaete-scute* complex. Cell *55*, 1061–1067.

Chandra, H. S. (1985). Sex determination: a hypothesis based on noncoding DNA. Proc. Natl. Acad. Sci. USA *82*, 1165–1169.

Cline, T. W. (1976). A sex-specific, temperature-sensitive maternal effect of the *daughterless* mutation of *Drosophila melanogaster*. Genetics *84*, 723–742.

Cline, T. W. (1978). Two closely linked mutations in *Drosophila melanogaster* that are lethal to opposite sexes and interact with *daughterless*. Genetics *90*, 683–698.

Cline, T. W. (1979). A male-specific lethal mutation in *Drosophila melanogaster* that transforms sex. Dev. Biol. *72*, 266–275.

Cline, T. W. (1980). Maternal and zygotic sex-specific gene interactions in *Drosophila melanogaster*. Genetics *96*, 903–926.

Cline, T. W. (1983). The interaction between *daughterless* and *Sex-lethal* in triploids: a lethal sex-transforming maternal effect linking sex determination and dosage compensation in *Drosophila melanogaster*. Dev. Biol. *95*, 260–274.

Cline, T. W. (1984). Autoregulatory functioning of a *Drosophila* gene product that establishes and maintains the sexually determined state. Genetics *107*, 231–277.

Cline, T. W. (1985). Primary events in the determination of sex in *Drosophila melanogaster*. In Origin and Evolution of Sex, H. O. Halvorson and A. Monroy, eds. (New York: Alan R. Liss), pp. 301–327.

Cline, T. W. (1986). A female-specific lesion in an X-linked positive regulator of the *Drosophila* sex determination gene, *Sex-lethal*. Genetics *113*, 641–663.

Cline, T. W. (1988). Evidence that *sisterless-a* and *sisterless-b* are two of several discrete "numerator elements" of the X:A sex determination signal in *Drosophila* that switch *Sxl* between two alternative stable expression states. Genetics *119*, 829–862.

Cronmiller, C., and Cline, T. W. (1986). The relationship of relative gene dose to the complex phenotype of the *daughterless* locus in *Drosophila*. Dev. Genet. *7*, 205–221.

Cronmiller, C., and Cline, T. W. (1987). The Drosophila sex determination gene *daughterless* has different functions in the germ line versus the soma. Cell *48*, 479–487.

Cronmiller, C., Schedl, P., and Cline, T. W. (1988). Molecular characterization of *daughterless*, a *Drosophila* sex determination gene with multiple roles in development. Genes Dev. *2*, 1666–1676.

Dambly-Chaudière, C., Ghysen, A., Jan, L. Y., and Jan, N. Y. (1988). The determination of sense organs in *Drosophila*: interaction of *scute* with *daughterless*. Roux's Arch. Dev. Biol. *197*, 419–423.

Davis, R. L., Weintraub, H., and Lassar, A. B. (1987). Expression of a single transfected cDNA converts fibroblasts to myoblasts. Cell *51*, 987–1000.

Davis, R. L., Cheng, P.-F., Lassar, A. B., and Weintraub, H. (1990). The MyoD DNA binding domain contains a recognition code for muscle-specific gene activation. Cell *60*, 733–746.

Dobzhansky, T., and Schultz, J. (1934). The distribution of sex factors in the X-chromosome of *Drosophila melanogaster*. J. Genet. *28*, 233–255.

Ellis, H. M., Spann, D. R., and Posakony, J. W. (1990). *extramacrochaetae*, a negative regulator of sensory organ development in Drosophila, defines a new class of helix-loop-helix proteins. Cell *61*, 27–38.

Erlich, H. A. (1989). PCR Technology (New York: Stockton Press).

Falk, R. (1963). A search for a gene control system in *Drosophila*. Am. Nat. *97*, 129–132.

Garcia-Alonso, L., and Garcia-Bellido, A. (1988). *extramacrochaetae*, a trans-acting gene of the *achaete-scute* complex of *Drosophila*, involved in cell communication. Roux's Arch. Dev. Biol. *197*, 328–338.

Garrell, J., and Modolell, J. (1990). The Drosophila *extramacrochaetae* locus, an antagonist of proneural genes that, like these genes, encodes a helix-loop-helix protein. Cell *61*, 39–48.

Gergen, J. P. (1987). Dosage compensation in *Drosophila*: evidence that *daughterless* and *Sex-lethal* control X chromosome activity at the blastoderm stage of embryogenesis. Genetics *117*, 477–485.

Halazonetis, T. D., Georgopoulos, K., Greenberg, M. E., and Leder, P. (1988). c-Jun dimerizes with itself and with c-Fos, forming complexes of different DNA binding affinities. Cell *55*, 917–924.

Hodgkin, J. (1983). Two types of sex determination in a nematode. Nature *304*, 267–268.

Hodgkin, J. (1987a). Primary sex determination in the nematode *C. elegans*. Development (Suppl.) *101*, 5–16.

Hodgkin, J. (1987b). Sex determination and dosage compensation in *Caenorhabditis elegans*. Annu. Rev. Genet. *21*, 133–154.

Hodgkin, J. (1989). Drosophila sex determination: a cascade of regulated splicing. Cell *56*, 905–906.

Hodgkin, J. (1990). Sex determination compared in *Drosophila* and *Caenorhabditis*. Nature *344*, 721–728.

Hooper, K. L., Parkhurst, S. M., and Ish-Horowicz, D. (1989). Spatial control of *hairy* protein expression during embryogenesis. Development *107*, 489–504.

Howard, K., and Ingham, P. (1986). Regulatory interactions between the segmentation genes *fushi tarazu*, *hairy*, and *engrailed* in the Drosophila blastoderm. Cell *44*, 949–957.

Ingham, P. W., Howard, K. R., and Ish-Horowicz, D. (1985a). Transcription pattern of the *Drosophila* segmentation gene *hairy*. Nature *318*, 439–445.

Ingham, P. W., Pinchin, S. M., Howard, K. R., and Ish-Horowicz, D. (1985b). Genetic analysis of the *hairy* locus in *Drosophila melanogaster*. Genetics *111*, 463–486.

Ish-Horowicz, D., and Pinchin, S. M. (1987). Pattern abnormalities induced by ectopic expression of the Drosophila gene *hairy* are associated with repression of *ftz* transcription. Cell *51*, 405–415.

Kouzarides, T., and Ziff, E. (1988). The role of the leucine zipper in the *fos-jun* interaction. Nature *336*, 646–651.

Lindsley, D. L., and Zimm, G. (1985). The genome of *Drosophila melanogaster*, Part 1: Genes A–K. Dros. Inf. Serv. *62*.

Lindsley, D. L., and Zimm, G. (1987). The genome of *Drosophila melanogaster*, Part 3: Rearrangements. Dros. Inf. Serv. *65*.

Lucchesi, J. C. (1978). Gene dosage compensation and the evolution of sex chromosomes. Science *202*, 711–716.

Lucchesi, J. C., and Skripsky, T. (1981). The link between dosage compensation and sex differentiation in *Drosophila melanogaster*. Chromosoma *82*, 217–227.

Lyon, M. F. (1972). X-chromosome inactivation and developmental patterns in mammals. Biol. Rev. *47*, 1–35.

Macdonald, P. M., and Struhl, G. (1986). A molecular gradient in early *Drosophila* embryos and its role in specifying body pattern. Nature *324*, 537–545.

Madl, J. E., and Herman, R. K. (1979). Polyploids and sex determination in *Caenorhabditis elegans*. Genetics *93*, 393–402.

Maine, E. M., Salz, H. K., Cline, T. W., and Schedl, P. (1985). The *Sex-lethal* gene of Drosophila: DNA alterations associated with sex-specific lethal mutations. Cell *43*, 521–529.

Mange, A. P., and Sandler, L. (1973). A note on the maternal effect mutants *daughterless* and *abnormal oocyte* in *Drosophila melanogaster*. Genetics *73*, 73–86.

McCoubrey, W. K., Nordstrom, K. D., and Meneely, P. M. (1988). Microinjected DNA from the X-chromosome affects sex determination in *Caenorhabditis elegans*. Science *242*, 1146–1151.

Moscoso del Prado, J., and Garcia-Bellido, A. (1984a). Cell interactions in the generation of chaetae pattern in *Drosophila*. Roux's Arch. Dev. Biol. *193*, 246–251.

Moscoso del Prado, J., and Garcia-Bellido, A. (1984b). Genetic regulation of the *achaete-scute* complex of *Drosophila melanogaster*. Roux's Arch. Dev. Biol. *193*, 242–245.

Murre, C., McCaw, P. S., and Baltimore, D. (1989a). A new DNA binding and dimerization motif in immunoglobulin enhancer binding, *daughterless*, *MyoD*, and *myc* proteins. Cell *56*, 777–783.

Murre, C., McCaw, P. S., Vaessin, H., Caudy, M., Jan, L. Y., Jan, Y. N., Cabrera, C. V., Buskin, J. N., Hauschka, S. D., Lassar, A. B., Weintraub, H., and Baltimore, D. (1989b). Interactions between heterologous helix-loop-helix proteins generate complexes that bind specifically to a common DNA sequence. Cell *58*, 537–544.

Nagoshi, R. N., McKeown, M., Burtis, K. C., Belote, J. M., and Baker,

B. S. (1988). The control of alternative splicing at genes regulating sexual differentiation in D. melanogaster. Cell *53*, 229–236.

Nakabeppu, Y., Ryder, K., and Nathans, D. (1988). DNA binding of three murine Jun proteins: stimulation by Fos. Cell *55*, 907–915.

Oliver, B., Perrimon, N., and Mahowald, A. P. (1988). Genetic evidence that the *sans-fille* locus is involved in *Drosophila* sex determination. Genetics *120*, 159–171.

Romani, S., Campuzano, S., and Modolell, J. (1987). The *achaete-scute* complex is expressed in neurogenic regions of *Drosophila* embryos. EMBO J. *6*, 2085–2092.

Romani, S., Campuzano, S., Macagno, E. R., and Modolell, J. (1989). Expression of *achaete* and *scute* genes in *Drosophila* imaginal discs and their function in sensory organ development. Genes Dev. *3*, 997–1007.

Rushlow, C. A., Hogan, A., Pinchin, S. M., Howe, K. M., Lardelli, M. T., and Ish-Horowicz, D. (1989). The *Drosophila hairy* protein acts in both segmentation and bristle patterning and shows homology to *N-myc*. EMBO J. *8*, 3095–3103.

Salz, H. K., Cline, T. W., and Schedl, P. (1987). Functional changes associated with structural alterations induced by mobilization of a P element inserted in the *Sex-lethal* gene of *Drosophila*. Genetics *117*, 221–231.

Salz, H. K., Maine, E. M., Keyes, L. N., Samuels, M. E., Cline, T. W., and Schedl, P. (1989). The *Drosophila* female-specific sex-determination gene, *Sex-lethal*, has stage-, tissue-, and sex-specific RNAs suggesting multiple modes of regulation. Genes Dev. *3*, 708–719.

Sánchez, L., and Nöthiger, R. (1983). Sex determination and dosage compensation in *Drosophila melanogaster*: production of male clones in XX females. EMBO J. *2*, 485–491.

Schröder, C., Tautz, D., Seifert, E., and Jäckle, H. (1988). Differential regulation of the two transcripts from the *Drosophila* gap segmentation gene *hunchback*. EMBO J. *7*, 2881–2887.

Spradling, A. C. (1986). P element-mediated transformation. In *Drosophila*, A Practical Approach, D. B. Roberts, ed. (Oxford: IRL Press), pp. 175–197.

Steinmann-Zwicky, M. (1988). Sex determination in *Drosophila*: the X-chromosomal gene *liz* is required for *Sxl* activity. EMBO J. *7*, 3889–3898.

Steinmann-Zwicky, M., Amrein, H., and Nöthiger, R. (1990). Genetic control of sex determination in *Drosophila*. Adv. Genet. *27*, 189–237.

Steller, H., and Pirrotta, V. (1985). A transposable P vector that confers G418 resistance to *Drosophila* larvae. EMBO J. *4*, 167–171.

Tapscott, S. J., Davis, R. L., Thayer, M. J., Cheng, P. F., Weintraub, H., and Lassar, A. B. (1988). *MyoD1*: a nuclear phosphoprotein requiring a *myc* homology region to convert fibroblasts to myoblasts. Science *242*, 405–411.

Tautz, D., Lehmann, R., Schnürch, H., Schuh, R., Seifert, E., Kienlin, A., Jones, K., and Jäckle, H. (1987). Finger protein of novel structure encoded by *hunchback*, a second member of the gap class of *Drosophila* segmentation genes. Nature *327*, 383–389.

Tautz, D., and Pfeifle, C. (1989). A non-radioactive in situ hybridization method for the localization of specific RNAs in *Drosophila* embryos reveals translational control of the segmentation gene *hunchback*. Chromosoma *98*, 81–85.

Torres, M., and Sánchez, L. (1989). The *scute (T4)* gene acts as a numerator element of the X:A signal that determines the state of activity of *Sex-lethal* in *Drosophila*. EMBO J. *8*, 3079–3086.

Vavra, S. H., and Carroll, S. B. (1989). The zygotic control of *Drosophila* pair-rule gene expression. I. A search for new pair-rule regulatory loci. Development *107*, 651–662.

Villares, R., and Cabrera, C. V. (1987). The *achaete-scute* gene complex of D. melanogaster: conserved domains in a subset of genes required for neurogenesis and their homology to *myc*. Cell *50*, 415–424.

Waring, G. L., and Pollack, J. C. (1987). Cloning and characterisation of a dispersed, multicopy, X chromosome sequence in *Drosophila melanogaster*. Proc. Natl. Acad. Sci. USA *84*, 2843–2847.

Wieschaus, E., and Nüsslein-Volhard, C. (1986). Looking at embryos. In Drosophila, A Practical Approach, D. B. Roberts, ed. (Oxford: IRl Press), pp. 199–226.